13.08.2025.

Module:-I

Research, Preperation, Planning, and Resources.

Research Methodology

- Framework or set of principles that guides how the research is conducted.
- It typically includes
  - → Research design (quantitative, qualitative (or) mixed)
  - → Data collection Methods (surveys, experiments, interviews, observations, etc).
  - → Sampling Techniques
  - → Data analysis procedure
  - → Justification for the chosen approach suitable for reports.
- In simple term → Road Map
  - → Answering the qns
    - what, why, how.

18.8.2025

Knowledge Economy,

stack holder,

Importance of knowledge economy.

Concept of Knowledge Economy

* The term knowledge economy was popularized by the world Bank and the OECD (organisation for Economic Cooperation and Development) to describe economies that rely on the production, distribution and use of knowledge as their primary source of growth.

* Knowledge economy is characterized by

- Heavy reliance on intellectual capital rather than just physical capital.
- Strong emphasis on innovation and research.
- Use of information and communication technologies to store, process and transmit knowledge
- Focus on lifelong learning, skills, and human development
- Global interconnectedness through the sharing of knowledge and research collaborations.

Definition:

The knowledge economy is one where knowledge becomes the key driver of economic activity, much like coal and steel were for the industrial revolution.

Role of Research in the Knowledge Economy:

* Research forms the backbone of the knowledge Economy.
* It is through research that new knowledge is generated, tested and applied for real world solutions.
* The connection b/w research and the knowledge economy can be understood as follows:

1. Knowledge Creation

- Research generates new knowledge in science, technology, social sciences and humanities

Eg: Discovery of renewable energy technologies.

2. Innovation and Application

- Research transforms theoritical knowledge into innovations products and services

Eg: Artificial intelligence research being applied in healthcare diagnostics.

3. Intellectual Capital Development:

- Research builds human resources with advanced skills and problem-solving abilities

Eg: Universities and R&D labs act as hubs of intellectual capital.

4. Policy and Decision Making:

- Evidence-based research guides governments and organisations to make better decisions.

Eg: Pandemic-related research guiding health policies worldwide.

5. Global Competitiveness:

- Countries with strong research infrastructure dominate the knowledge economy.

- India and china are rapidly investing in research to strengthen their global standing.

Characteristics of Research in a Knowledge Economy

* Interdisciplinary : complex problems require collaboration across multiple fields.

* Technology-driven : Use of big data, machine learning and advanced software in research

* Globalized : International research collaborations and networks are common.

* Innovation-oriented : Focus not only on knowledge generation but also on commercial applications.

* Continuous Learning : knowledge is constantly updated, requiring lifelong learning and adaptability.

Benifits of Research in the Knowledge Economy:

The integration of research and the knowledge economy brings several benifits:

1. Economic Growth:

- Research-driven innovations lead to new industries and job creations.

Eg: IT revolution in India has significantly contributed to GDP.

2. Social Development

- Research improves healthcare, education and communication system.

Eg: Medical research has enabled breakthroughs like Vaccines for COVID-19.

3. Sustainability

- Research enables development of eco-friendly technologies and sustainable practices.

Eg: Development of solar, wind and hydrogen energy systems.

4. Global competitiveness:

- Nations that invest in research and development (R&D) remain leaders in technology and economy.

Eg: South Korea spends over 4% of GDP on R&D, making it a global tech leader.

Challenges in Building a Research-oriented Knowledge Economy:

while research is crucial for the knowledge economy, there are certain challenges:

- High costs of R&D : Advanced research requires significant financial investment.
- Knowledge Divide : Developing countries may lag behind due to lack of research infrastructure
- Intellectual Property Rights (IPR) : Issues of knowledge ownership and access can create barriers.
- Brain Drain : Talented Researchers often migrate to developed countries for better opportunities.
- Ethical Issues : In areas like AI, biotechnology, and genetic engineering, ethical concerns must be carefully addressed.

Key stakeholders:

1. Researchers and Academics
2. Universities and Research Institutions
3. Government and Policymakers
4. Funding Agencies
5. Industry and corporate sector

6. Students and Early Career Researchers

7. Society and Communities

8. International organisations

9. Media and knowledge Dissemination platforms.

Stake holder Interrelationship in knowledge Economy.

The stakeholders do not act independently; rather they form a knowledge ecosystem:

- Governments fund universities and research institutes.
- Universities collaborate with industry for technology transfer
- Industry commercializes research for society.
- Society benefits, provides feedback and influences government policies.

This creates a cycle of knowledge → innovation → application → feedback → new research.

21.08.2025

Types of Research

1. Basic (Pure) Research:

Basic or pure research is conducted to improve the understanding of fundamental principles and theories. It is not directed towards solving any immediate practical problem rather focuses on expanding scientific knowledge.

Purpose:

- Understand natural phenomena or underlying mechanisms
- Lay the groundwork for future applied research and innovations.

Characteristics:

- Driven by curiosity and intellectual interest
- Focused on why and how questions.

2. Applied Research:

Applied research is focused on solving specific, practical problems, using existing theories and knowledge. It bridges the gap b/w theoritical knowledge and practical implementation directly contributing to innovation and development.

Purpose:

Improve existing processes, materials or technologies.

characteristics:

Focused on "How to" questions.

significance:

converts basic research into products, innovations etc.

3. Descriptive Research:

It aims to accurately describe the characteristics of a particular individual group, situation or phenomenon. It answers the questions, "what, when, where, and how".

Purpose:

Decision making, and future research

characteristics:

Utilizes quantitative and qualitative data collection.

4. Analytical Research

Purpose:

- To analyze existing datasets or research findings.
- To identify patterns, correlations or trends for better understanding or decision making.

characteristics:

- Heavily relies on data analytics, statistics and logical reasonings.

- Provides a foundation of predictive modeling and forecasting.

Significance

- Helps make informed conclusions and evidence-based recommendations.
- converts raw data into meaningful insights for decision making or further research.

5. Exploratory Research

Purpose:

- Understand the nature of an unknown problem
- Generate hypotheses and guide future detailed studies

characteristics

- Flexible, open-ended and qualitative in nature
- Involves brainstorming.
- Does not provide final answers.

significance

- Helps formulate research problems, hypotheses or strategies for deeper investigation.

6. Explaratory Research:

- Also known as casual research.
  - understands the reasons behind a phenomenon.

Purpose :

- Understand why and how events or behavior occur
- Establish relationships b/w independent and dependent variables.

characteristics

- Quantitative in nature with controlled experiments.

7. Experimental Research :

→ able to find relationship b/w cause and effect

→ scientific approach to observe effect on other variables under controlled conditions.

Purpose

* establish cause and effect relationships

characteristics

* conducted in labs or controlled field environments.
* Follows a systematic procedure.
  1) Formulate a hypotheses
  2) Design and setup experiments
  3) Collect and analyse data

4) Draw conclusions and validate hypotheses

* Requires careful control of external variables to avoid bias

Significance:

* Provides highly reliable results due to controlled conditions

* Essential for developing new materials medical treatments and engineering innovations.

8. Quantitative Research:

- focusing on collecting and analyzing numerical data

- Predicts the outcomes

Purpose:

* measures and quantify variables for objective evolution

* Predicts outcomes or trends using statistical and computational methods

Characteristics

* objective and data driven

* uses large sample size for reliability and reproducibility

* employs mathematical models, statistical tests and charts to interpret results

Significance :

* Support data-driven decision making
* Enables trend forecasting and hypothesis testing in engg., healthcare and social sciences.

9. Qualitative Research :

- explores deeper meanings, motivations and experiences
- deals with non-numerical data.

Purpose :

* Gain in-depth understanding of human behavior perceptions
* identifying patterns, themes and narratives rather than numerical trends.

characteristics

* subjective and interpretive in nature
* usually involves small sample sizes but provides rich insights
* Data collected is textual, visual or verbal not numeric

Significance :

* provides context meaning and depth to complement quantitative studies
* useful in problem exploration social sciences and behavioral.

10. Action Research :

- To find solutions to immediate problems faced by individuals or organizations

Purpose.

* improve practices, processes or performances in real world settings

* involve stakeholders in the research and solution development process.

Characteristics

* collaborative and participatory research.

11) Cross-sectional Research :

Purpose

* snapshot of current conditions

* identify correlations and patterns without studying changes over time.

Characteristics

* usually qualitative

* cost effective and time efficient

* cannot establish cause and effect relationships

Significance

* provides quick.

12) <u>Longitudinal Research</u>:

- studies a subject or sample over an extended period

<u>Purpose</u>

* Track growth, progress or degradation over time

* identifying long term effects casual patterns and prodictions.

<u>29.08.25</u>.

Critical and Creative Thinking

<u>Critical Thinking</u>

- systematic process of analyzing, evaluating and interpreting information to form a well reasoned judgement

- objective, logical and evidance based enabling researchers to identify flows in argument, spot biases and question assumptions

<u>Purpose</u>:

→ Evaluate the validity, reliability and relevance of informations

→ Identify flows, gaps and biases in studies or arguments

→ Make informed conclusions - based on evidence rather than assumptions.

→ critical thinkers are skeptical (doubtful) but open-minded relying on reasoning in data rather than opinions.

- In research, critical thinking is crucial for evaluating literature, designing methodology, interpreting results, and drawing valid conclusions.

Eg:

critically reviewing conflicting studies on hydrogen storage allows a researcher to assess the strength of evidence, spot gaps and propose better approaches

- Analysing the cause and effect, based on that the researcher come to a conclusion.

characteristics:

→ Analytical - Breaks complex problems into smaller parts

→ Objective - Avoids personal bias or emotional influence

→ Logical and systematic - follows clear reasoning steps.

→ skeptical but open-minded - Questions claims but consider new evidence

→ Evidence-oriented - Requires data and facts to support conclusions.

Applications in Research:

→ Literature Review: Assessing the credibility of existing studies

→ Methodology Design: choosing the most valid and reliable approach.

→ Data interpretation: Avoiding misinterpretation or overgeneralization of results

→ Data conclusions: Ensuring inferences are supported by evidence.

Examples:

→ critically reviewing conflicting studies on hydrogen storage to identify which experiments are most reliable.

→ Detecting statistical bias or misinterpretation in a paper on composite materials

→ Evaluating the credibility of AI predictions for material performance before practical applications.

Significance

→ Ensures accuracy, validity and credibility in research outcomes

→ Helps researchers avoid errors, false assumptions and weak arguments.

Creative Thinking :

Creative thinking is the ability to look at problems or situations from fresh perspective and to propose innotive solutions

- It often involves thinking "out of box" using imagination and combining ideas in new ways.

Purpose :

→ Generate novel hypotheses and innovative solutions

→ Develop unique experimental designs or interdisciplinary approaches

→ overcome research challenges by thinking beyond conventional methods.

Characteristics :

→ Innovative : proposes original solutions or perspectives

→ curious and open-minded : Explores diverse ideas and approaches

→ Flexible : Adapts and modifies approaches when needed

→ Risk-Taking : willing to try unconventional method

→ Integrative : Connects concepts across discipline.

- In research, creativity helps in developing novel, hypotheses, designing experiments, or using unconventional methods.

- A creative researcher might combine techniques from multiple fields (Eg: AI with material sciences) to develop new models or technologies. while critical thinking ensures rigor and validity, creative thinking drives originality and discover.

Both are complementary and essential for high-quality research.

Applications :

→ Designing novel experimental setups for new materials testing

→ combining AI machine learning and materialscience to creative predictive models.

→ Developing in new data visualisations techniques to identify hidden trends.

Examples :

→ Integrating nanotechnology with hydrogen storage research to develop high-capacity alloys

→ Proposing an interdisciplinary research method combined.

→

Significances:

→ Drives innovations, originality and discovery in research.

Comparison between critical thinking and creative thinking

Aspect	Critical Thinking	Creative Thinking
1) Definition	A systematic process of analyzing, evaluating & judging information to make logical decisions	A process of generating new original and innovative ideas by thinking beyond existing boundaries.
2) Focus	Accuracy, clarity, logic, evidence and rationality	Imagination, originality, novelty and flexibility
3) Goal	To make the best possible decision or judgement	To think of new possibilities and innovative solutions
4) Approach	convergent - narrows down to the best possible solution	Divergent - expands to multiple possible solutions
5) Skills involved	Reasoning, questioning, comparing, evaluating, Problem-solving	Brainstorming, imagining experimentary, exploring alternative.
6) Nature of thinking	Rational, objective, skeptical, Disciplined	intutive, flexible, openminded freeflowing.

Aspects	Critical Thinking	Creative Thinking
7) outcome	Produces well-reasoned judgements, solutions or decisions	produces innovative concepts designs, and new possibilities
8) Example	Evaluating whether a research paper has valid methodology and data	Designing new approach to wireless power transfer for UAVs.

Generating a Research Topic:

→ one of the most challenging steps in research process

→ A Good topic is relevent, original, specific and feasible within the available time and resources.

→ The process begins by identifying a broad area of interest, such as renewable energy or health care and then narrowing it down through literature review, expert discussion(or) real-world observations.

→ Researchers should explore gaps in existing knowledge, unresolved issues or areas with practical significance.

→ Tools like mind mapping, brainstorming and SWOT analysis can help to generate and refine ideas.

→ Reviewing academic journals, attending seminars (or) analysing trends in the industry can also provide direction.

→ For instance if a reseacher is interested in magnesium based hydrogen storage, they may explore under researched parameters like the role of synthesis pressure on desorption time, eventually formulating a focused researchable question.

→ A well defined research topic leads to clear objectives guides the research design and ensures the study adds value to the existing body of knowledge.

Journal - Microgrid

Identify the Area of Interest

↓

Preliminary Literature Review

↓

Identifying Gaps / Problems / Trends

↓

Brain storm possible topic (SWOT)

↓

Evaluate Feasibility

↓

Refine and Narrow down to topic

↓

Formulate Research Problem / question

↓

Define Research objectives

↓

Finalize the Research Topic

02.09.25.

1) Identifying Broad Research Question :

- Design of Questionnare :

Criteria

→ choose a research area

→ Literature Review → identifying research gap

→ Identify the problem statement → keep it broad.

→ keep it in broader one (Don't focus on variable or metrics yet)

2. Good Research objective

→ It should be SMART.

S - specific

M - Measurable

A - Acheirable

R - Relerant

T - Time Bound.

→ Road map for entire research process

→ It should be understandable and simple sentence.

3. Types of objectives

Generic specific.

Importance :

- foundation of the study
- knowledge gap or society need

sources

- Literature Review
- Industrial Problem
- Personal Interest

characteristics

- open ended, relevant and researchable
- scope.

Linking Questions to the objective.

1) Quantitative → Results to numbers
2) Qualitative → emotions

Likert scale

- Its like feed back. 1 2 3 4 5

4) Select Question Types

- close ended - MCQs, Yes or no, Likert scale qns.
- open-ended - Detailed manner, opinions
- Ranking - Rating, prioritizing, factors
- Ensure clarity and flow